

A screened OPERA dataset for km-scale extreme precipitation

From the v1 patch set to the v4 screen: what was found, built and decided

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Outline

Why the dataset was rebuilt

The raw archive

v2: cleaning, leak-free splits, events

Validation against rain gauges

v3: repair instead of reject

The POT threshold study exposes three radar failures

The v4 screen

Phase 0: calibration

Gallery review (7 October)

Isolated high values

Open items and next steps

The v1 patch set could not support the tail claims

- ▶ **Splits leak.** Patches were shuffled individually over 2023-08 to 2024-10 on the same 79 tile locations. For 100% of test patches (including $\max \geq 89 \text{ mm h}^{-1}$) a train patch exists at the same timestamp and on the same tile within $\pm 1 \text{ h}$. Test scores are interpolation scores.
- ▶ **The DEM channel was N–S mirrored.** The DEM GeoTIFF is north-first and the precipitation stores south-first; both were sliced with the same index. 20/20 test patches match the mirrored slice (e.g. correlation -0.32 with the true terrain).
- ▶ **The tail was censored and holed.** Preprocessing set every value $> 150 \text{ mm h}^{-1}$ to zero, not to 150. The 435 original days had also been capped at 150 mm h^{-1} upstream.
- ▶ **The tail was contaminated.** On the test split the share of patches whose mass sits in a few pixels ($\bar{x}/x_{\max} < 10^{-3}$) rises with the threshold: 25.9% at 31, 29.3% at 89 mm h^{-1} . Real cells wet more area as they intensify.
- ▶ **Only 15 months.** Too little to hold out unseen extremes.

Fixes: the DEM is looked up by patch position (`src.data.geo.dem_patch`); everything else needs a new dataset built from the full archive.

The OPERA archive on disk

- ▶ Source: EUMETNET open radar data, anonymous S3 bucket `openradar-archive` (`scripts/data/fetch_opera_archive.py`). Rain rate + quality index (QIND), ODIM HDF5, LAEA grid 2200×1900 , 2 km, 15 min.
- ▶ **4,683 days complete**, 2012-09-04 to 2025-10-22 (the other 114 calendar days have no data), 1.2 TB. 2025-10-23 to 2026-09-22 (334 days) stopped on the user quota.
- ▶ The 435 original 2023–24 stores (capped at 150 mm h^{-1} , no QIND) were replaced by archive refetches; they agree 100% below 149 mm h^{-1} (footprint IoU 0.9999).
- ▶ **The raw field is dirty.** On 2018-06-15, 52 of 96 composites have a maximum above 500 mm h^{-1} ; the peak is $13,072 \text{ mm h}^{-1}$. Offenders are isolated pixels.
- ▶ **Product breaks** to watch when pooling years: late 2015 (blockage correction, cloud mask), 2017-09-29 (log $\rightarrow$ linear compositing), **2024-07-05** (ODYSSEY $\rightarrow$ NIMBUS).

ODYSSEY vs NIMBUS: the same days, two products

ODYSSEY composites the scans of a 15-min window; NIMBUS takes the lowest-elevation scan at the nominal time. Paired comparison on the 101 days held in both (2,407 frames, 97,682 wet tiles):

quantity	NIMBUS / ODYSSEY
wet area ≥ 0.1 / 1 mm h^{-1}	0.71 / 0.99
exceedance ≥ 10 / 31 / 89 mm h^{-1}	1.36 / 1.44 / 2.4
wet-pixel p50 / p90 / p99	1.44 / 1.38 / 1.45
perimeter at $10\text{--}31 \text{ mm h}^{-1}$	≈ 1.8
spectral power at 128 / 32 / 8 / 4 km	1.25 / 1.42 / 3.1 / 6.2
+15 min correlation	0.41 $\rightarrow$ 0.26

NIMBUS fields are more intense, rougher and more fragmented at small scales. **Decision (DECISIONS §18):** train/val/test are ODYSSEY only; NIMBUS is a separate product-shift test split (`nimbus`, `events_nimbus`).

v2 pipeline (`scripts/dataset_v2/`)

1. **Scan.** Per-year clutter climatology; every frame cleaned (`src/data/cleaning.py`: drizzle floor, static-clutter and spike *repair*, no zeroing above 150 mm h^{-1}); every fully covered 128×128 tile described.
2. **Splits** (`make_splits.py`):
 - ▶ whole ISO weeks drawn per calendar month, 1-day buffers $\Rightarrow$ train is ≥ 2 days from val/test;
 - ▶ 26 catalogued events (ECMWF, MeteoSwiss, DWD, ESSL, AEMET, Météo-France studies) forced into test;
 - ▶ tiles rejected if unphysical ($> 500 \text{ mm h}^{-1}$ after repair) or an RLAN ray;
 - ▶ stratified sample of 3M patches with per-row inclusion weights.
3. **Store**, gamma targets, `check_datasets.py`.

Subsets: `full_*`, `light_*` (25%), `extremes_*` ($\max \geq 31 \text{ mm h}^{-1}$), `events_test` (every tile of every event).

Built 2026-10-01/02: train 1,977,908 / val 251,667 / test 282,684 / nimbus 487,743 patches on 3,168 / 404 / 463 / 801 days. All checks passed. The log1p scaler moves from 5.02 to ≈ 6.2 because nothing is zeroed any more.

Audit flags and consistently bad radars

- ▶ **Temporal support:** a cell $\geq 10 \text{ mm h}^{-1}$ with no echo $\geq 1 \text{ mm h}^{-1}$ within 30 km at $t \pm 15 \text{ min}$.
- ▶ **Range rings:** invisible in single frames, but circles in the per-year $\geq 31 \text{ mm h}^{-1}$ frequency. Stevns and Sindal (DK, $\sim 234 \text{ km}$, 2013–17), Ikaalinen and Vimpeli (FI, 2012); none after the 2017-09-29 compositing change.
- ▶ **Bad radars** (whole archive, relative to the 5 nearest radars): 14 of 261 consistently bad, e.g. Bollène (clutter, 15/15 years), Teolo ($> 500 \text{ mm h}^{-1}$), Ängelholm, Stevns, Sindal (unsupported maxima), Berlin (ring 2013–15).
- ▶ The first ranking flagged 204 of 246 radars. Three bugs: a zero 95th percentile used as outlier bound; duplicate current/archive database entries splitting a site; 16 sites without ODIM code dropped by a groupby.

Gauge data and design

- ▶ Independent truth: 10-min gauges from DWD (1,454) and MeteoSwiss SwissMetNet (281); 12.1M station-frames on 4,968 days.
- ▶ Gauge QC on the gauges alone: fault codes (>50 mm/10 min), stuck runs, isolated bursts with no rain within 30 km. 230 of 84.7M wet values masked.
- ▶ Two questions, by raw intensity bin and per rule:
 - ▶ *presence*: did the gauge see rain (≥ 1 mm h $^{-1}$) under the pixel?
 - ▶ *values*: exceedance ratio $\rho(u) = N(\text{radar} \geq u) / N(\text{gauge} \geq u)$ over the same station-frames, no conditioning on either side. A flat ρ means the tail has the right shape; its level mixes point-vs-2 km and instantaneous-vs-10-min offsets.
- ▶ The radar frame matches the gauge interval 10–20 min later (log-rate correlation 0.29–0.47).

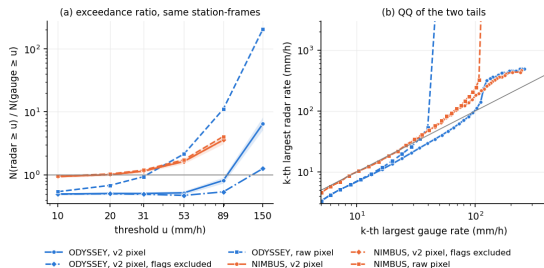
Presence: which rules remove rain?

Share of pixels with gauge $\geq 1 \text{ mm h}^{-1}$, by raw intensity:

class	[10,31)	[31,89)	[89,150)	[150,500)	≥ 500
untouched	0.91	0.92	0.91	0.90	–
no temporal support	0.01	0.01	0.00	0.00	0.01
on range ring	0.82	0.50	0.00	0.00	0.00
spike repaired	0.13	0.24	0.24	0.26	0.00
tile rejected: unphysical	0.81	0.74	0.55	0.10	0.05
tile rejected: ray	0.73	0.57	0.25	0.00	0.06

- ▶ Temporal support is the cleanest rule; rings carry real rain below 31 mm h^{-1} , none from 89.
- ▶ **Tile rejection discards real rain:** a rejected tile is usually a real storm with one bad pixel. $\Rightarrow$ repair the pixel, keep the tile (v3).
- ▶ QIND does not discriminate (ODYSSEY AUC 0.26, inverted; NIMBUS 0.53).

Values: is the tail calibrated?



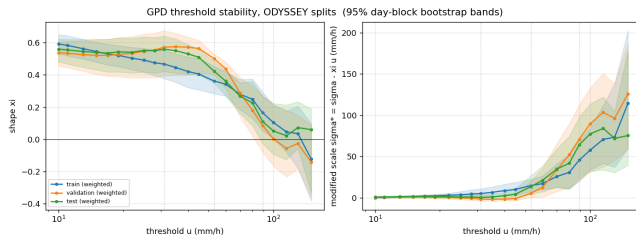
$\rho(u)$	31	89	150
ODYSSEY raw	0.93	11.0	204
ODYSSEY v2	0.51	0.82	6.5
flags excluded	0.49	0.54	1.26
NIMBUS v2	1.14	3.58	31

- ▶ ODYSSEY exceeds u about half as often as the gauges at every level up to 53 mm h^{-1} (89 with flags excluded): the tail *shape* matches, and $u = 31$ sits on the calibrated part.
- ▶ NIMBUS is heavier than the gauges from $\approx 31 \text{ mm h}^{-1}$, and no rule touches it.
- ▶ At gauge $\geq 30 \text{ mm h}^{-1}$ the cleaned radar 3×3 max is $\geq 10 \text{ mm h}^{-1}$ in 71% of cases; median radar/gauge 0.45.

v3 rebuild (done 2026-10-05)

- ▶ **Repairs replace tile rejection.** Footprints ($\geq 150 \text{ mm h}^{-1}$ regions around a >500 core), ray components, ring pixels $\geq 89 \text{ mm h}^{-1}$, and cells without temporal support are lowered to the median of their outer ring.
- ▶ **Validated on the same 12.1M gauge pairs.** ODYSSEY ρ at 150 mm h^{-1} : $6.46 \text{ (v2)} \rightarrow 1.47 [1.12, 1.92]$. Under repaired pixels the gauge saw $\geq 10 \text{ mm h}^{-1}$ in 5.3% of cases (untouched: 44.6%). NIMBUS unchanged.
- ▶ Ring repair starts at 89 mm h^{-1} , not 31: rings at $31\text{--}89 \text{ mm h}^{-1}$ held real rain (38.5% gauge $\geq 10 \text{ mm h}^{-1}$).
- ▶ **Event set fixed:** the 2013 Central European floods and the Andreas hailstorms were training days in v2; Emilia-Romagna recovered with shifted windows.
- ▶ **Built** in 33 h on 8 workers: train 1,962,057 / val 252,809 / test 297,460 / nimbus 487,673; events_test 36,166 tiles of 25 events, events_nimbus 5,249 (Boris, Valencia DANA). All checks passed; min gap between splits 2 days.

No threshold is stable on v3

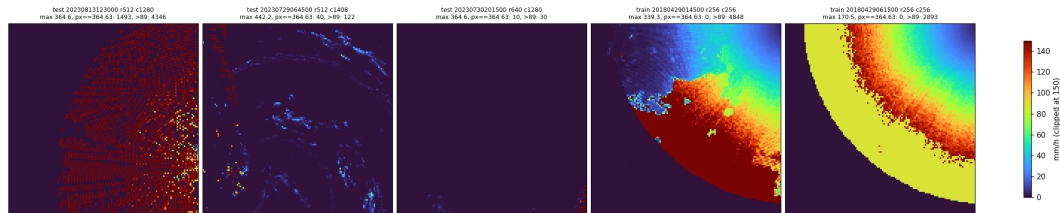


GPD fits over $u = 10\text{--}150 \text{ mm h}^{-1}$,
day-block bootstrap ($B = 200$).

ξ falls monotonically: train $+0.47$ at
31, $+0.17$ at 89, -0.12 at 150.

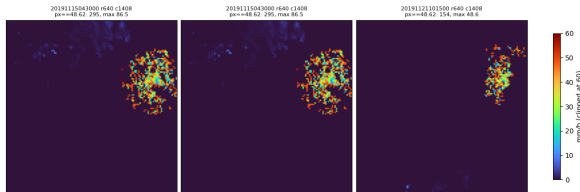
Sub-asymptotic shape, or artefacts?
The top 10 train days hold 37% of the
exceedances of 89 mm h^{-1} , so look at
them.

Two failures shape the v3 tail



- ▶ **$364.63 \text{ mm h}^{-1} = 64.0 \text{ dBZ}$** under $Z = 200R^{1.6}$: a reflectivity ceiling of the Valjevo (Serbia) area on 80 rain days of 2023; 12% of all test pixels $>89 \text{ mm h}^{-1}$.
- ▶ **Torrejón de Velasco (Madrid), 2018-04-29**: a whole radar disk at a smooth range-dependent $60\text{--}340 \text{ mm h}^{-1}$; 16% of all train exceedances of 89 mm h^{-1} . The v3 repair lowered it to its ring median, which is the same failing radar: a flat 88.27 mm h^{-1} plateau.
- ▶ Removing only these tiles (797 of 1.07M train) flattens $\xi(u)$: $+0.34$ to $+0.37$ over $31\text{--}89 \text{ mm h}^{-1}$.

A third failure, found by a repeated-value test



Real rain on the 0.01 mm h^{-1} ODYSSEY grid almost never repeats one value: per tail tile, median 1 and 99.9th percentile 24 repeats of one value $\geq 31 \text{ mm h}^{-1}$.

≥ 50 repeats flagged 0.069% of train tail tiles and found **$48.62 \text{ mm h}^{-1} = 50.0 \text{ dBZ}$** , Oradea (Romania), tile r640 c1408, 7 days in November 2019.

Why v3 missed all three: each v3 rule assumes an artefact that is small, local or transient. These are large, smooth or persistent.

Five rules, each keyed on an error signature (DECISIONS §19)

1. **Repeated value per tile** (≥ 50 repeats of one value $\geq 31 \text{ mm h}^{-1}$; on NIMBUS counted in excess of the adjacent ladder levels).
2. **Ceiling table** per radar area and year. A ceiling pixel is censored, not measured: ≥ 5 per tile-frame $\Rightarrow$ reject; 1–4 $\Rightarrow$ repair like a spike.
3. **Radar-disk failure** per radar and frame: wet fraction, share of log-rate variance explained by range alone, jump at the area boundary and the maximum-range circle.
4. **Guarded repair**: `boundary_fill` refuses components above a size limit; the repeated-value test runs after every repair.
5. **Radar-day ranking** against each radar's own seasonal level and its 5 nearest radars; exclusions from a list reviewed by eye.

Policy (DECISIONS §17): reject clear errors only, keep imperfections, no labelled set. None of the rules keys on intensity alone; every rule reports its rejection rate by intensity bin, because real extremes look anomalous too.

When is the dataset ready? A gate fixed in advance (DECISIONS §20)

Ready when one more cleaning step would not change any number we report.

1. **Known failures caught**, kept in a regression catalogue.
2. **Residual contamination measured**: 100 random test tiles with $\max \geq 89 \text{ mm h}^{-1}$ looked at; 95% upper bound on the artefact rate $\leq 5\%$.
3. **Evaluation reference stable**: dropping the 20 most influential non-event days moves no observation-side table number outside its day-block bootstrap interval.
4. **Real extremes survive**: $< 0.5\%$ rejected per event; no rule's rejection rate climbs steeply with intensity; gauge ρ no worse than v3.
5. **Residual drift in $\xi(u)$ explained** (truncation at 500 mm h^{-1} , mixing, sub-asymptotic threshold) with a truncated-GPD fit.

No cleaning by tail fit: selecting data by its fit to the GPD and then using the GPD as evaluation reference would be circular. The fit points at candidates; rules key on a ceiling value, a range-only field, a repeated value.

Build plan

phase	content	status
1	code: <code>src/data/radar_screen.py</code> , guarded repair, tests	done
0	calibration on known failures, 27 events, 100 random days	run 2 passed
2	radar pass over all days, ceiling table, radar-day review list	next (~ 9 h)
3	tile scan with the v4 chain	(~ 12 h)
4	store, gamma targets, checks	(~ 20 h)
5	audit = the \$20 gate	

Decisions are taken at split time from per-tile columns, so a threshold can change without a rescan. Retraining starts only after the gate.

Run 1 failed on the ceiling rule

First rule 2: a value counted $> 50\times$ the median of its occupied ± 0.5 dB neighbours.

- ▶ **3,256 radar-year “ceilings”** where a real ceiling is one value. Event gate failed entirely through rule 2: valencia_dana 100%, ce_floods 93%, boris 93% of wet tiles rejected.
- ▶ Cause: a radar-year mixes values on coarse ladders (dBZ steps of 0.4–3 dB) with sparse off-ladder values (counts 1–4). The median of the neighbours is then ≈ 1 and every ladder level passes.
Example: Málaga 2024, 10.28 mm h^{-1} counted 8,059 times among neighbours counted 1–4.

New comparison: the count of a value against the *largest* count of any other value within ± 4 dB (wider than the coarsest ladder step, ≈ 3 dB). Rain counts fall with intensity, so on a ladder the ratio is $\lesssim 1$; a ceiling piles up above its neighbours.

- ▶ Calibration (199 days, 23,868 values counted ≥ 50): the 18 rows of the two known ceilings score 9.9–89; all other values have 99.9th percentile 1.96. $\Rightarrow$ min_ratio: 5 (provisional).

Run 2: gates

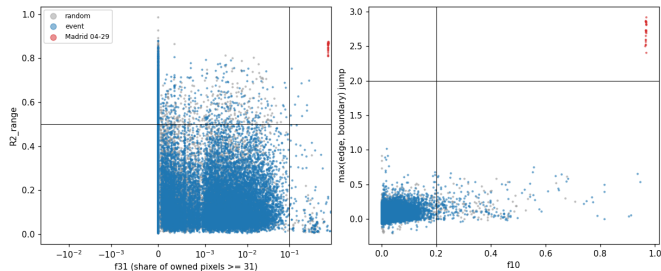
Ceiling table: 21 rows. 364.63 mm h^{-1} on 17 radar-years (Serbia, Croatia, Thessaloniki, Brindisi), 48.62 mm h^{-1} Oradea 2019, and $11.53 \text{ mm h}^{-1} = 40.0 \text{ dBZ}$ on three of the same Balkan radars in 2023.

Gate 1: known failures	tiles	caught
Madrid 2018-04-29, $\geq 10\%$ Madrid-owned, $\max \geq 31$	101	96
364.63 (2023) and 48.62 (2019) in the ceiling table	2	2
48.62 days, tile r640 c1408, $\max \geq 31$	411	310

Gate 2 passed: worst event 0.14% rejected (chauxdefonds, 1 tile, rule 1); 24 of 25 events 0%.
Random days: rejection $\leq 0.014\%$ below 150 mm h^{-1} , 0.74% at $\geq 150 \text{ mm h}^{-1}$ (Valjevo-area tiles).
All 110 tiles with $\max > 500 \text{ mm h}^{-1}$ are rejected.

Regression: with v4 switched off, 2018-06-15 reproduces the v3 table.

Rule 3: the radar-disk failure separates cleanly



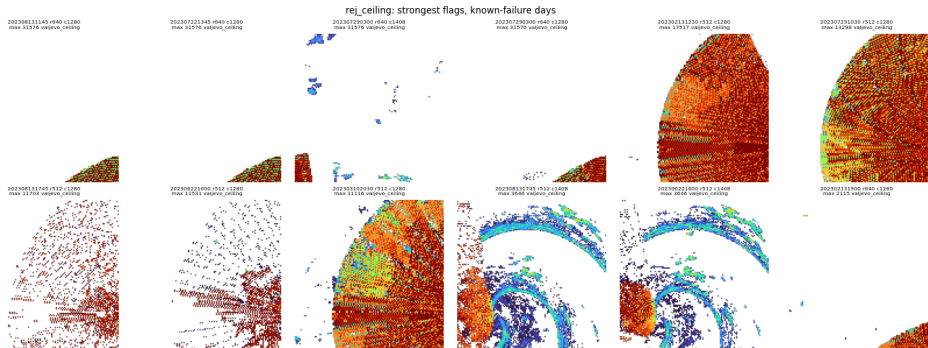
Per radar and frame (wet frames, wet fraction ≥ 0.2):

Madrid 04-29 frames: share $\geq 31 \text{ mm h}^{-1}$ 0.82, R^2 of log-rate on range 0.86, jump 2.4–2.9.

Random and event frames: edge jump 99.9th percentile 0.10–0.14, boundary jump 0.38–0.48; largest jump of either 1.02.

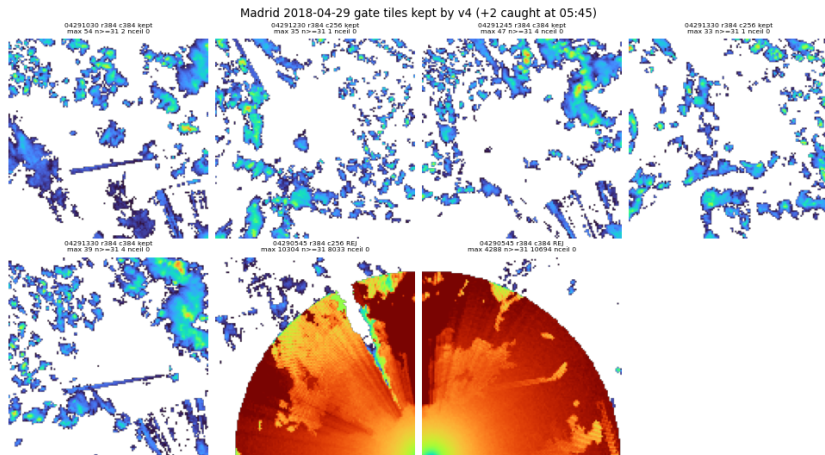
22 other flagged radar-frames on 13 radar-days, mostly the 2023 Balkan radars.

The rule galleries are dominated by one failure



The galleries show each rule's 12 strongest flags by tile max. All of them come from the 2023 Balkan failure; the “random / event” panels fall inside its window (02-13 to 08-13), so they are true positives too. Every panel is visibly broken: spokes, saturated sectors, speckle. The open Gate-1 questions need galleries targeted on them.

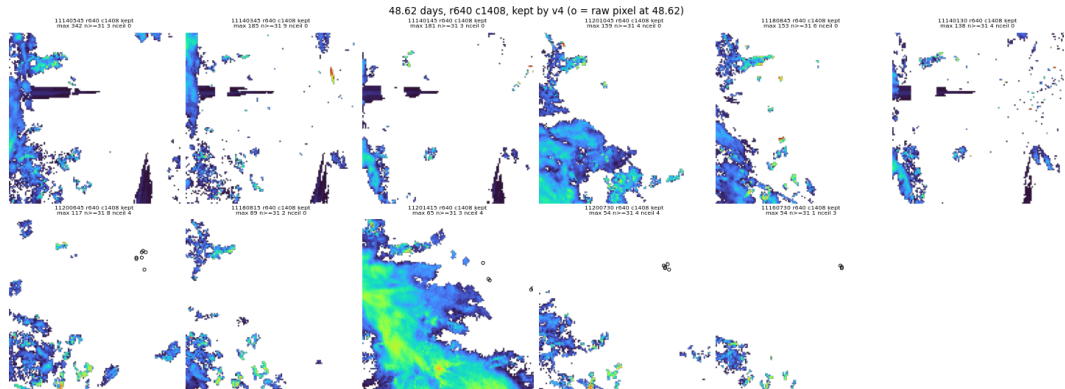
Madrid: the five kept tiles are outside the failure



All 96 gate tiles in the failure window (00–06 h) are rejected. The 5 kept tiles are from 10:30–13:30, max 33–54 mm h⁻¹, scattered convection; the two right-hand panels show the same tiles at 05:45.

Closed.

48.62: the ceiling is handled, a second failure is not

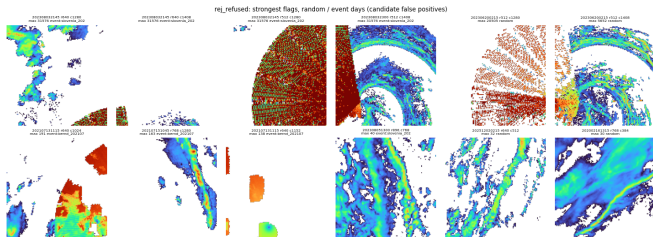


Kept frames carry 0–4 raw 48.62 pixels (circles), repaired as designed. But their maxima ($138\text{--}342\text{ mm h}^{-1}$) are clusters of 2–9 pixels whose 5×5 median (10 km) is 0 mm h^{-1} . The repair can copy such values: $5,524.7 \rightarrow 153.4\text{ mm h}^{-1}$ from an equally isolated neighbour.

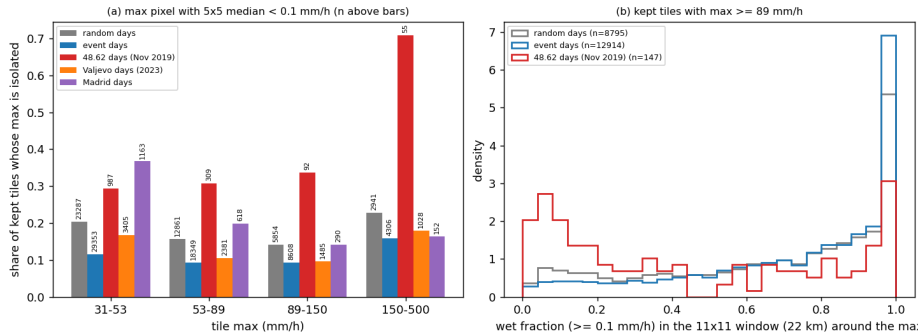
Two smaller findings

11.53 mm h⁻¹ (40.0 dBZ), 2023. 398 pixels in total on three radars (grthe, rsfrg, itbri). Its largest days (02-14, 06-11, 07-12, 07-22, 07-29) are all 364.63 failure days, so it is probably the same fault; the calibration set over-samples those days, so the link is weak. It only triggers repairs (no tile below 89 mm h⁻¹ is rejected for a ceiling). Kept flagged.

Refused repairs on real rain. Three tiles rejected because the guard refused a repair look like ordinary frontal bands (max 30–40 mm h⁻¹). Rate on random days $\leq 1.4 \times 10^{-4}$ per intensity bin; still to check what was refused.



Isolated high values: how often, and where?



Phase-0 days, 118,627 tiles with $\max \geq 31 \text{ mm h}^{-1}$, v4-cleaned field (scripts/dataset_v2/isolated_v4.py). A pixel is isolated when the median of its 5×5 window (10 km) is $< 0.1 \text{ mm h}^{-1}$ and the window is $\geq 50\%$ covered.

Isolation alone is not a clear-error signature

- ▶ On the 48.62 days the max is isolated far more often: 71% of kept tiles at $150\text{--}500\text{ mm h}^{-1}$, against 16% on event days and 23% on random days.
- ▶ **But the event days show it too:** 9–16% of kept event tiles have an isolated max, in every intensity bin.
- ▶ The strictest cut tried (wet fraction < 0.05 in the 11×11 window, 22 km) still flags 1.8% of event tiles with $\text{max} \geq 89\text{ mm h}^{-1}$, against 15.6% on the 48.62 days and 2.7% on random days. Gate 2 allows $< 0.5\%$ per event.
- ▶ Whether the isolated event maxima are small real cells or artefacts inside event windows is not known; they have not been looked at.

Reading: a per-tile isolation rule would remove real storms at several times the gate rate (DECISIONS §17, safeguard 1). The 48.62 days stand out by *rate*, which is a radar-day property. The natural home is rule 5 (radar-day ranking): the share of isolated maxima per radar and day, against the radar's own level and its neighbours. That would have to be added to the Phase-2 radar pass before it runs.

A by-product: $486.2 \text{ mm h}^{-1} = 66.0 \text{ dBZ}$

- ▶ 55 tiles have a max of exactly 486.2 mm h^{-1} ; no other value between 470 and 500 mm h^{-1} is the max of more than 3 tiles.
- ▶ 486.2 mm h^{-1} is 66.0 dBZ under $Z = 200R^{1.6}$, the same kind of round reflectivity value as 364.63 (64.0 dBZ) and 48.62 (50.0 dBZ).
- ▶ 36 of the 55 tiles are on event days, 15 random, 4 Valjevo; it is also the v3 maximum of both NIMBUS events (Boris, Valencia DANA).
- ▶ It is not in the ceiling table: per radar-year it is too rare for `min_count`: 50 or does not pile up $5\times$ above its neighbours.

Interpretation open: a product-level cap (values clipped at 66 dBZ, i.e. censored hail cores) is consistent with this, but has not been checked. If it is a cap, the pixels are real heavy rain with an unknown true value, and the tail above $\approx 486 \text{ mm h}^{-1}$ is censored, not truncated at 500. The Phase-2 radar-year histograms will show whether it is radar-specific.

Open items

- ▶ **Isolated maxima:** no tile-level rule. Decide whether to add the share of isolated maxima per radar-day to rule 5; it must go into the radar pass before Phase 2.
- ▶ **486.2 mm h⁻¹:** product cap or radar fault? Read from the Phase-2 histograms.
- ▶ Commit the rule-2 change; launch Phase 2 (radar pass, ~9 h, all 8 cores).
- ▶ Review the radar-day list by eye; record exclusions in `quality_v4.yaml`.
- ▶ Phases 3–5 (~32 h of compute), then the §20 gate.
- ▶ Re-derive the POT threshold on v4 (`pot_threshold.py --version v4`), with a truncated GPD.
- ▶ NIMBUS tail excess over the gauges: test with DWD 1-min records.
- ▶ Re-run the ring-repair gauge validation at 89 mm h⁻¹ (it was run at 31).
- ▶ Archive fetch 2025-10-23 to 2026-09-22 (334 days), blocked by quota.
- ▶ Talk to MeteoSwiss (D. Nerini, L. Moret) on filters, plausibility bounds and o.o.d. test design.

Known limits of the dataset

- ▶ Tiles need 100% radar coverage: coastal and network-edge extremes (Mediterranean) are under-represented; HyMeX IOP16 is absent from the archive.
- ▶ Radar rates underestimate intense rain (at gauge $\geq 30 \text{ mm h}^{-1}$ the median radar/gauge ratio is 0.45; Valencia: 28.6 mm h^{-1} radar pixel vs a 184.6 mm h^{-1} gauge-hour at Turís).
- ▶ NIMBUS evaluation carries a product penalty (small-scale variance, persistence, heavier tail) unrelated to extremeness; report it against a same-product reference.
- ▶ The data are truncated at 500 mm h^{-1} after repair; a truncation alone makes ξ fall at high u .
- ▶ No labelled set: precision and recall of the rules on real extremes are bounded by the gallery review, the per-bin rates and the gauges, not measured.